8. (a)

Fuelling the Future

The main fuel used as a petrol substitute for cars is bioethanol. Bioethanol fuel is mainly produced by the fermentation of sugar.

The main sources of the sugar required to produce bioethanol are corn, maize and wheat. These crops absorb carbon dioxide from the atmosphere.

The use of pure bioethanol in car engines is only possible if the engines are designed for that purpose. For this reason, bioethanol-petrol mixtures called 'blends' are used.

Bioethanol-petrol mixtures have "E" numbers that describe the percentage of bioethanol in the mixture by volume. See **Diagram 1**.

Pure bioethanol is hard to vaporise. This makes it difficult to start a car when the weather is cold, which is why the fuels are almost always a bioethanol-petrol mixture.

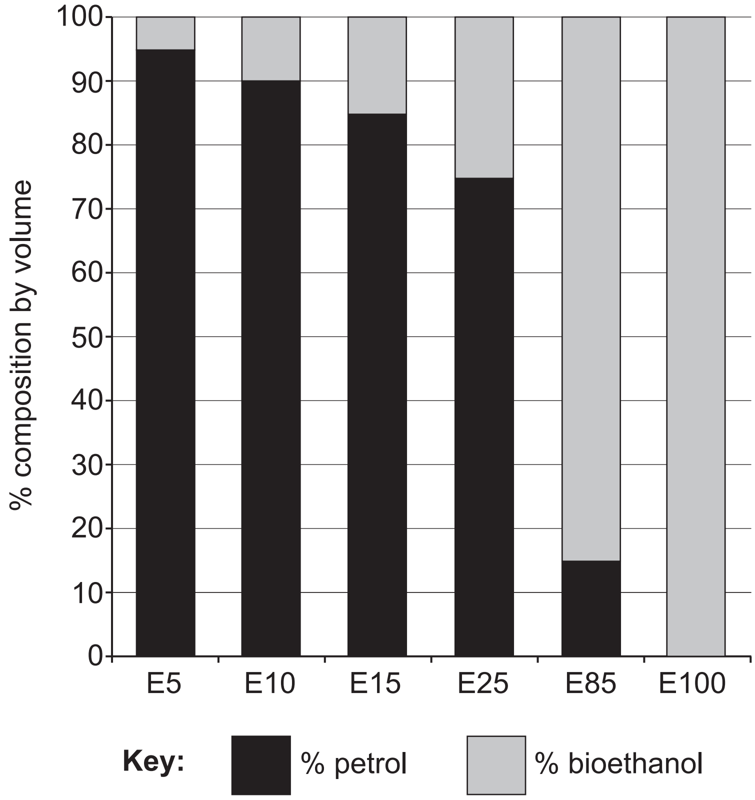

Diagram 1

Diagram 2 shows three properties of some fuels.

Property	Petrol	E10	E20	E30	E40	E60	Bioethanol
Density (kg/m^3)	747.4	750.8	760.5	778.2	779.2	781.2	789.0
Flash point (°C)	−65	−40	−20	−15	−13.5	−1.0	12.5
Energy value (MJ/kg)	44.40	44.22	42.08	40.48	38.50	35.84	29.78

Diagram 2

The flash point is the lowest temperature at which the fuel ignites.

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Some people claim that bioethanol is 'carbon neutral' because the plants that are the source of the bioethanol absorb carbon dioxide as they grow. This compensates for the carbon dioxide released when bioethanol is burned. However, carbon dioxide is also emitted during the construction of the fermentation factory, during fermentation and when bioethanol is transported around the world.

In some parts of the world, large areas of forest have been cleared and burned to develop the bioethanol industry. Deforestation to make space for crops has a devastating effect on the environment because it drastically reduces the number of trees that can capture carbon dioxide emissions. The amount of land needed to grow the biomass material is considered the main drawback of bioethanol as a fuel.

(i) Tick (✓) the box next to the ratio of bioethanol : petrol in the E20 fuel blend. [1]

80% bioethanol : 20% petrol ☐

20% bioethanol : 20% petrol ☐

20% bioethanol : 80% petrol ☐

(ii) Tick (✓) the box next to the statement that describes the effect of increasing the percentage of bioethanol in a fuel blend. [1]

the energy value decreases ☐

the density decreases ☐

the flash point decreases ☐

(iii) Give **one** reason that undermines the claim that burning bioethanol is 'carbon neutral'. [1]

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(b) The molecular formula for ethanol is C_2H_5OH.

(i) Draw the structure of ethanol. [1]

(ii) When ethanol is exposed to air it slowly forms ethanoic acid.

When ethanoic acid reacts with magnesium carbonate, a salt is formed.

I. Give the name of the salt. [1]

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II. The negative ion in the salt has the formula CH_3COO^-.

<u>Underline</u> the correct formula of the salt. [1]

$MgCH_3COO$ **Mg_2CH_3COO** **$Mg_2(CH_3COO)_2$** **$Mg(CH_3COO)_2$**

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(c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**.

Her results are recorded in the table.

	Add dilute HCl	Add $BaCl_2$(aq)	Add NaOH(aq)	Add $AgNO_3$(aq)
A	gas given off turns limewater milky		pungent smelling gas given off that turns damp red litmus paper blue	
B	no reaction	white precipitate forms	green precipitate forms	no reaction
C	no reaction	no reaction	blue precipitate forms	white precipitate forms

Draw a line from the unknown compound to its correct chemical name. [3]

Unknown compound	**Chemical name**
	copper(II) chloride
	iron(II) chloride
A	ammonium sulfate
B	iron(III) chloride
C	copper(II) carbonate
	ammonium carbonate
	iron(II) sulfate

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